UML Relationships Explained

This document explains common **UML relationships between classes/objects**, ordered from **weakest to strongest coupling**.

Overview

All relationships describe how **Class A depends on Class B**, but the *strength and nature* of that dependency differ.

Strength order:

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Dependency < Association < Aggregation < Composition < Inheritance
```

Memory shortcut:

```
uses → knows → has → owns → is
```

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1. Dependency (Weakest)

Definition: Class **A uses class B temporarily**.

Key Points: - Short-lived relationship - No ownership - Changes in B may affect A - Usually via method parameters or local variables

Example: - `ReportService` uses `Printer` inside a method

Mental model:

> "I use you, but I don’t keep you."

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2. Association

Definition: Object **A knows object B**.

Key Points: - A holds a reference to B - B exists independently - Long-term relationship

Example: - `Teacher` knows `Student`

Mental model:

> "I know you."

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3. Aggregation

Definition: Object **A has B**, but **B can live without A**.

Key Points: - Whole–part relationship - Weak ownership - B may be shared - Represented by a *hollow diamond* in UML

Example: - `Team` has `Player` - Player still exists if Team is removed

Mental model:

> "I have you, but you’re not mine forever."

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4. Composition

Definition: Object **A owns B completely** and manages its lifecycle.

Key Points: - Strong ownership - A creates and destroys B - B cannot exist without A - Represented by a *filled diamond* in UML

Example: - `House` has `Room` - No house → no room

Mental model:

> "You live and die with me."

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5. Implementation

Definition: Class **A depends on the concrete implementation of B**.

Key Points: - Tighter coupling than interface-based dependency - Changes in B’s implementation directly affect A

Example: - Using a concrete class instead of an interface

Mental model:

> "I depend on how you are built."

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6. Inheritance (Strongest)

Definition: Class **A is a type of B**.

Key Points: - Strongest coupling - A inherits behavior and structure from B - Changes in B strongly affect A

Example: - `Dog extends Animal`

Mental model:

> "I am a kind of you."

Final Summary Table

Relationship	Meaning	Ownership	Strength
Dependency	uses	❌No	Weakest
Association	knows	❌No	Weak
Aggregation	has	⚠️Weak	Medium
Composition	owns	✅Strong	Strong
Inheritance	is	✅Full	Strongest

Key Takeaway

Choose the **weakest relationship possible** to reduce coupling and make your system easier to maintain and extend.